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Lua Integration Showcase

OmniLaTeX

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Demonstrating LuaLaTeX's embedded Lua engine

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1 Inline Lua Execution

The `\directlua` command executes Lua code and prints the result of `tex.print()` calls directly into the document.

$2 + 2 = 4$

Lua can access the current date and time:

2 Lua Functions

Define reusable Lua functions in a `\directlua` block, then call them later.

Hello, World! Welcome to LuaLaTeX. Hello, OmniLaTeX User! Welcome to LuaLaTeX.

3 Table Manipulation

Lua tables are powerful data structures. Here we create a table of programming languages, sort it, and typeset it as an itemized list.

- Go
- Haskell
- Lua
- Python
- Rust

A word-count table built from a sentence:

4 String Processing

Lua's pattern-matching library is ideal for text transformations.

Extracting all numbers from a string:

5 Mathematical Computation

Compute the first 12 Fibonacci numbers using a Lua function:

n	1	2	3	4	5	6
$F(n)$	1	1	2	3	5	8

n	7	8	9	10	11	12
$F(n)$	13	21	34	55	89	144

6 Custom Lua-Backed Commands

Define LaTeX commands whose implementations live entirely in Lua.

Long date: ISO date:

= =

7 Practical Example: Auto-Generated CSV Table

A genuinely useful pattern — parse CSV data with Lua and render it as a $\LaTeX$ table with automatic alignment detection.

8 TeX-Lua Interaction

Pass data from $\LaTeX$ into Lua and back:

Project name: , version , built with .